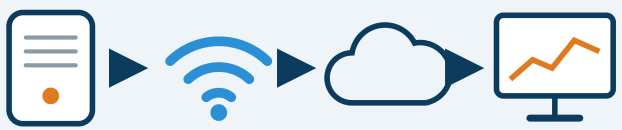
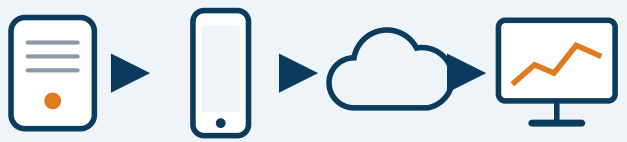
ParticularMatter: Use Cases

Four ways to deploy a KongQi node: indoor or outdoor, fixed or mobile

Federico Dallo (ISP-CNR)
Horizon EU healthRiskADAPT · particularmatter.org/setup.html

2026-10-19

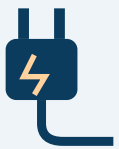
Pick your scenario

	FIXED	MOBILE
INDOOR	 **Wi-Fi → cloud** live dashboard	 **phone hotspot → cloud** + phone GPS (Strava)
OUTDOOR	 **GPS + SD card** or Wi-Fi in range → as indoor	 **GPS + SD card** keep sky in view

INDOOR · FIXED

Home Wi-Fi

1



Power on

`KongQi`

2



Join hotspot

`ESP32_Setup`

`particularmatter`

3



Pick home Wi-Fi

`2.4 GHz only`

4



Get one-time key

`dashboard.`

`particularmatter.org`

5



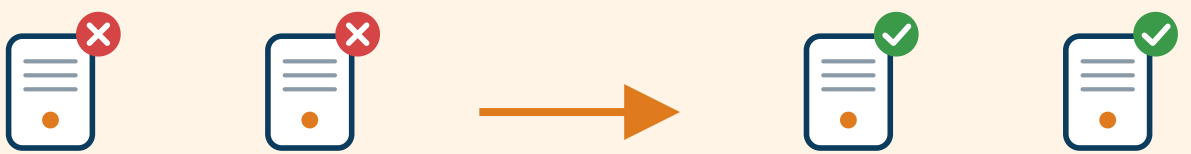
Paste key → live

`Node settings`

`192.168.4.1`

More than one device?

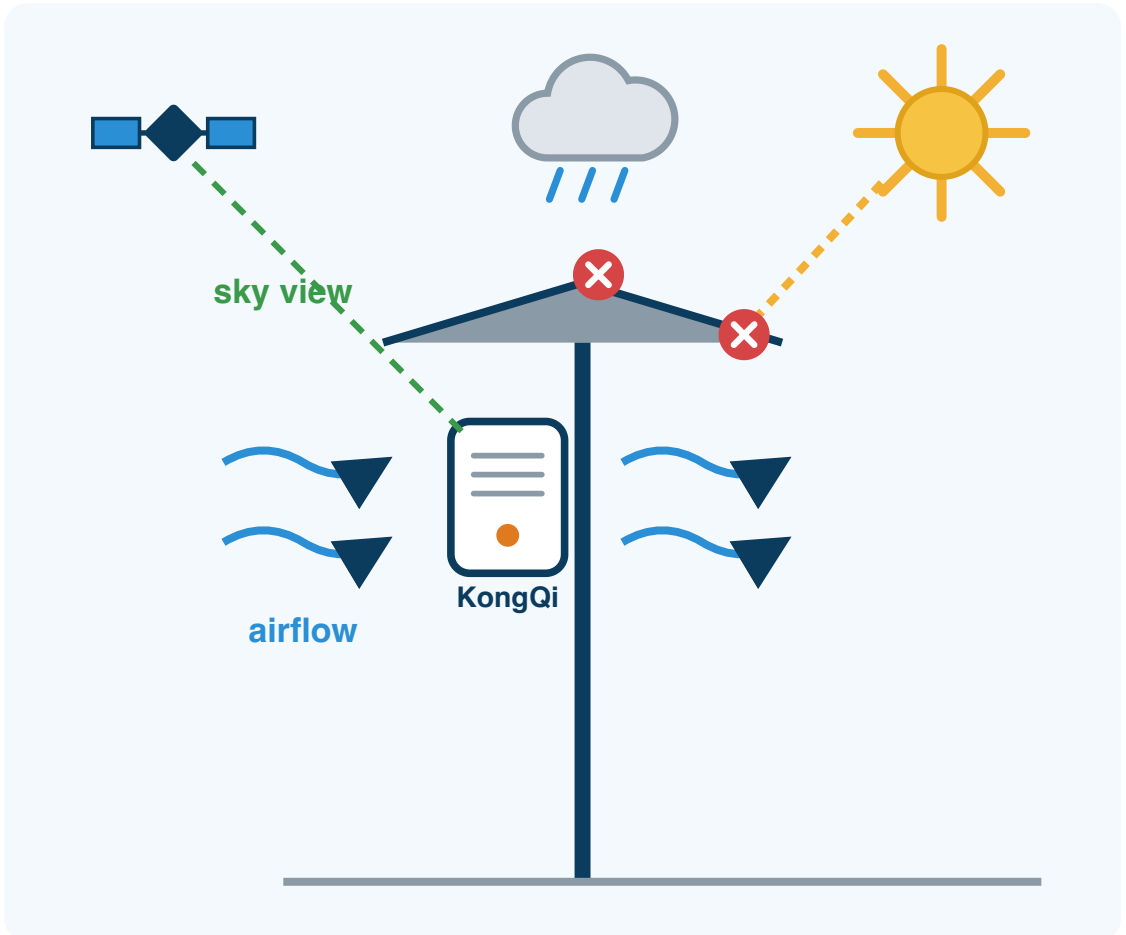
Rename each hotspot
(Node settings)



`ESP32_Setup` `ESP32_Setup` `KongQi-01` `KongQi-02`

OUTDOOR · FIXED

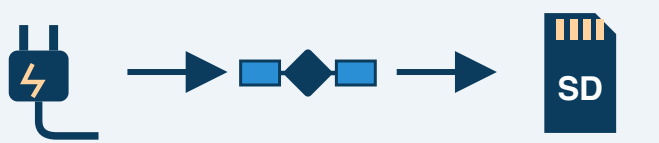
GPS + SD card, or Wi-Fi



- Good air circulation
- Clear view of the sky
- Direct sunlight
- Rain

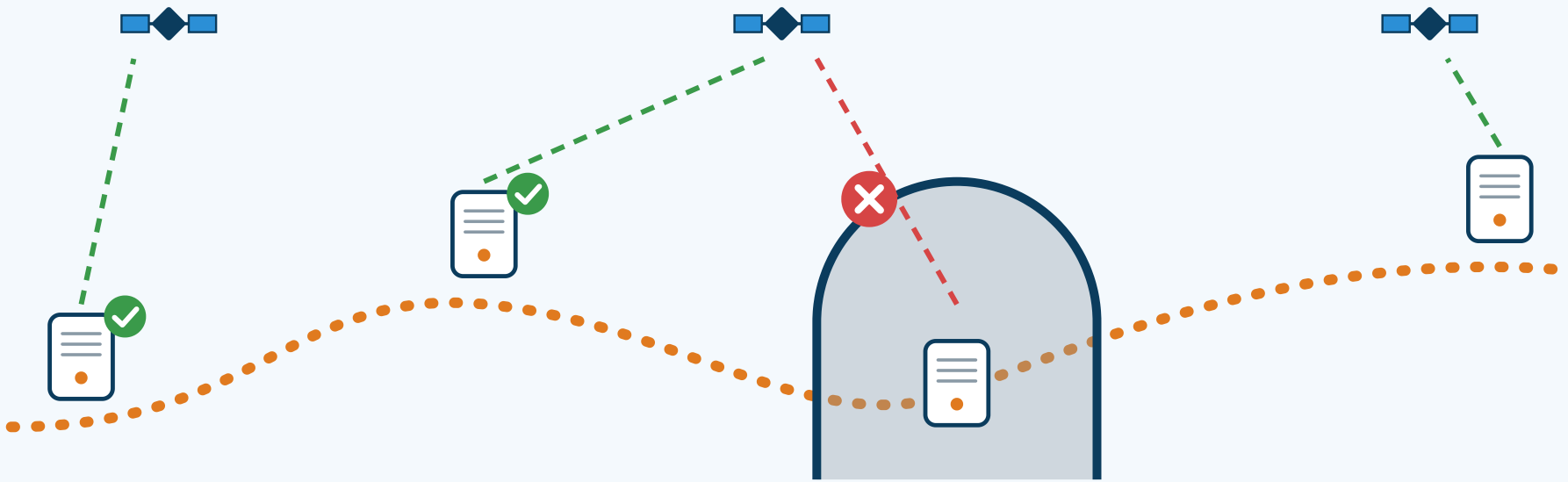
Wi-Fi in range?
Set it up like the indoor node

No Wi-Fi: logs stay on the device



OUTDOOR · MOBILE

Keep the sky in view



SD

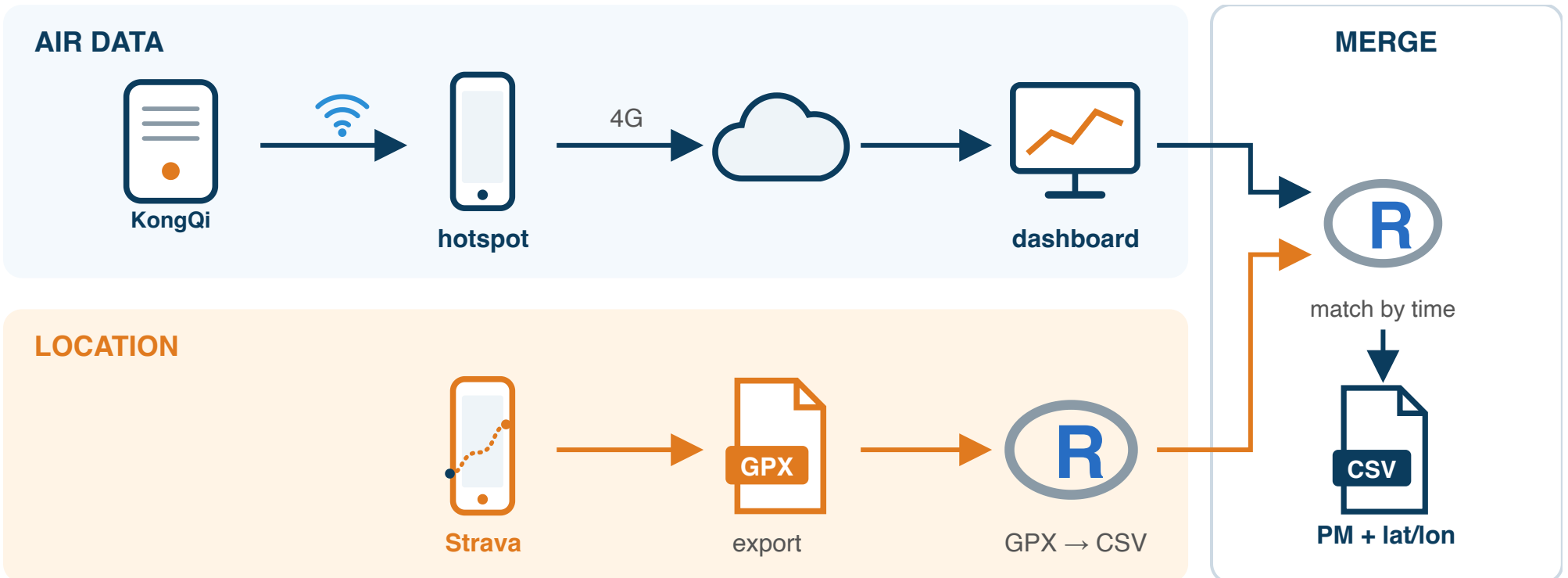
Powered & logging to SD

Check GPS fix before moving

Same setup as fixed outdoor

INDOOR · MOBILE

Phone = Wi-Fi + GPS



Indoors the phone GPS beats the device GPS → the phone provides internet and position

GPX: Strava activity on the web → ⋯ → **Export GPX** · R scripts: GPX → CSV · match with device data

More information

Full setup guide

particularmatter.org/setup.html

Dashboard

dashboard.particularmatter.org

Questions? federico.dallo@cnr.it